Wireshark for SCADA

Industrial Control Systems · SCADA Hub

COURSE TOPICS

1. **Introduction to Wireshark:** Three-pane interface, dissector architecture, ICS protocol support, and the difference between tshark and dumpcap.
2. **Capture Techniques:** SPAN ports, network taps, promiscuous mode, ring buffers, pcap/pcapng formats, and remote capture over SSH.
3. **Display Filters:** BPF capture filters, Wireshark display filter syntax, SCADA-specific expressions, colorization rules, and Follow TCP Stream.
4. **Protocol Dissectors:** Dissector mechanics, Decode As for non-standard ports, Wireshark profiles, and Statistics tools for traffic pattern analysis.
5. **Modbus Analysis:** MBAP header fields, transaction ID tracking, write command identification, and exception response detection in captures.
6. **DNP3 Analysis:** Frame structure, unsolicited response identification, IIN bit monitoring, and Secure Authentication v5 handshake verification.
7. **OPC UA Analysis:** Session establishment, SecureChannel negotiation, security mode verification, and subscription traffic profiling from captures.
8. **Security Analysis:** Traffic baseline establishment, anomaly detection, scanning identification, and incident response capture procedures.
9. **Advanced Techniques:** tshark scripting, SSH remote capture, Lua dissectors for proprietary protocols, and packet replay with tcpreplay.
10. **Protocol Lab:** Modbus poll overload diagnosis, Exception Code 02 trace, OPC UA security audit, rogue master detection, tshark alert script.

LEARNING OUTCOMES

- Capture ICS protocol traffic using SPAN ports and network taps without network disruption.
- Write Wireshark display filters targeting Modbus, DNP3, and OPC UA traffic patterns.
- Verify OPC UA security mode configuration from a packet-level session capture.
- Identify event buffer overflow, exception responses, and authentication handshakes in captures.
- Automate protocol monitoring with tshark scripts for recurring or unattended analysis tasks.

Certification Relevance

- **GICSP (GIAC):** ICS network analysis and incident response are core exam domains; Wireshark proficiency is assumed by the exam objectives.
- **GRID (GIAC):** ICS defender certification; network forensics and anomaly detection are heavily tested.
- **CEH (EC-Council):** Packet analysis and protocol investigation appear in the network reconnaissance modules.
- **CompTIA CySA+:** Network traffic analysis in the threat detection domain.

Next Steps

1. Complete Wireshark as the final course in the 8-guide sequence — protocol knowledge from earlier courses makes the captures readable.
2. Configure a home lab SPAN port or network tap and capture live Modbus or OPC UA traffic from a simulator.
3. Register at giac.org for GICSP; review exam objectives and map completed courses to each domain.
4. GICSP requires no specific work experience; the exam is available at Pearson VUE testing centers worldwide.